



Decision support

Impact of technology spillovers and subsidies on innovation consortia dynamics

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ABSTRACT

In innovation consortia, governments and enterprises view technology spillovers and subsidies differently, which reduces the effectiveness of policies and undermines cooperation. This work investigates the dynamics of cooperation among leading enterprises (LEs) and small/medium enterprises (SMEs), using an evolutionary game model with intrinsic, extrinsic and mixed strategies. We show that there may exist more than one evolutionary equilibrium of enterprise cooperation under multi-strategy choices. A high subsidy may induce enterprises to speculate and reach a suboptimal evolutionary equilibrium. Furthermore, our findings indicate that SMEs are particularly sensitive to changes in subsidy levels, income distribution ratios, and project income, with their speculative behavior exhibiting a lagging effect. Also, increasing the income distribution ratio of LEs can promote active cooperation between the groups and reduce the speculative tendency of SMEs. This occurs because it magnifies the effect of technology spillover on the income of SMEs and enhances the income of the leading enterprises. Our work improves understanding of cooperative factors and provides new recommendations for government and leading enterprises to improve cooperation and managerial performance.

1. Introduction

As research and development (R&D) activities become more complex, collaborative innovation among multiple entities has become increasingly necessary. Through resource integration and sharing achievements, enterprises progressively enhance their R&D performance using collaborative innovation (Lin et al., 2020). To support this activity, the Chinese government has designed a collaborative innovation model of *innovation consortia*, spearheaded by leading enterprises (LEs), to formulate related policies. The LEs are large enterprises that have amassed expertise in advanced technologies and hold a dominant position within their respective industries. In this model, the government assumes a guiding and supervisory role but allows enterprises substantial autonomy in their R&D activities.

However, governments and enterprises view technology spillovers and subsidies differently, which reduces the effectiveness of policies and undermines cooperation, and even makes enterprises treat the cooperation negatively. For example, governments aim to encourage active cooperation using low subsidies, since high subsidies have been shown to limit innovation (Liu et al., 2019). Conversely, selfish enterprises prefer high subsidies, which they may internalize as additional

profits, disregarding their incentive effects on cooperation. Also, governments hope to enhance technology spillover between industrial sectors to promote development (Lee et al., 2022). Conversely, LEs hope to curtail technology spillovers in order to maintain their competitive advantages (Guisado-González et al., 2018). Under enterprise-led cooperation, these cognitive differences are magnified, affecting the stability of cooperation, especially in innovation consortia that include both LEs and small and medium-sized enterprises (SMEs).

A typical innovation consortium is an aluminum manufacturing consortium led by an LE in China, which includes over 40 participating entities. The consortium aims to pool resources and accelerate innovation efficiency through collaborative improvement. The sources of funding for the consortium, as outlined in its charter, include collective fundraising, government grants, and business income. However, competitive relationships among enterprises lead to concerns regarding technology spillover within the consortium, potentially undermining cooperation. A common form of spillover damage is that enterprises enter cooperation by winning bids at low prices and profit from technological spillovers and the appropriation of outcomes. To mitigate this,

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the consortium strengthens incentives, penalties, and other mechanisms to forestall breakdowns in cooperation.

A related issue is the cooperation motivation of enterprises. Positive behavior in cooperation is driven by intrinsic motivations (IM), while negative behavior is driven by extrinsic motivations (EM). Several studies have explored the strategies of positive and negative behavior in cooperation among enterprises, which can be seen as two extreme cases of collaboration (Zan et al., 2021). In practice, however, it is common for enterprises to cooperate while also engaging in opportunistic behavior. This is described as cooperation strategies driven by mixed motivations (MM). Opportunistic behavior can lead to losses in the benefits of cooperating enterprises and even result in project failure. In a capsule intelligent manufacturing initiative, an LE established a consortium with multiple small and medium-sized enterprises (SMEs) to secure government-funded project approval. The LE employed an MM strategy, overstating the number of equipment deliveries to obtain excess government subsidies. Additionally, the LE delayed payments to SMEs to enforce timely delivery and prevent technology leakage resulting in significant financial losses for SME partners. These practices not only compromised project timelines and output quality but also triggered legal disputes, highlighting governance deficiencies in consortium-based innovation projects. Hence, providing a better understanding of factors such as technology spillovers and subsidies for cooperation motivation is valuable for the maintenance of R&D cooperation.

Existing studies have underscored the substantial impact of subsidies, technology spillovers, and project income on the stability of the collaborative innovation system (Ho, 2023; Su et al., 2020). Notably, there exists an inverted U-shaped relationship between the subsidy amount and various technological innovation indicators (Liu et al., 2019; Song et al., 2022). While technology spillover contributes to positive externalities, it may also induce harmful cooperative behavior among enterprises, such as free-riding (Li, Yi, & Zhao, 2020). Due to the dynamic nature of the collaborative innovation system and its susceptibility to various influencing factors, conventional research predominantly relies on evolutionary game theory and methodology to simulate scenarios wherein decision makers demonstrate bounded rationality in cooperative engagements (Liu et al., 2017; Zan et al., 2021). Enterprises engage in collaboration driven by either intrinsic or extrinsic motivation, influenced by various factors. Thus, cooperation strategy is typically defined as either positive or negative cooperation in classic studies. However, situations where these two motivations coexist have received less investigation. Current studies fail to address the cooperation stability problem under multiple strategy options. Only a small number of studies have investigated the evolutionary game of multiple strategies, for example (Gu et al., 2023), which focuses on the impact of cost and penalty factors on bike-sharing systems, whereas our work focuses on the impact of revenue and technology spillover changes on cooperation system. The context of our work is innovation consortia comprised of LEs and SMEs.

This paper studies the following questions:

1. Which cooperation statuses among enterprises in innovation consortia allow evolutionary stability under multi-strategy selection?
2. How do LEs and SMEs respond differently to changes in subsidies, technology spillovers, and income levels?
3. What measures can governments and LEs formulate to enhance income distribution and reward systems within innovation consortia, thereby improving synergy and cooperative performance?

We develop a two-participant and multi-strategy evolutionary game model, which studies the evolving dynamics of cooperation among enterprises under the influence of multiple factors and strategies. In this model, each participant has three strategy choices: intrinsic motivation, extrinsic motivation, and mixed motivation. Through simulation, we

evaluate the collective impact of factors such as project income, technology spillover, and subsidy levels on cooperation. Our findings can be summarized as follows:

1. Active cooperation by both sides, free-riding of SMEs, and joint speculation by both LEs and SMEs, can generate evolutionary equilibria.
2. LEs are more sensitive to technology spillovers, while SMEs are more sensitive to changes in other income factors.
3. Increasing the proportion of income distribution to LEs can stimulate SMEs to engage in active cooperation by adopting intrinsic motivations.

Our research makes several contributions to the management of innovation consortia. First, we develop a multi-strategy evolutionary game model to study the dynamics of cooperation between enterprises, which extends the binary classification of strategies as positive and negative. This expands the application of evolutionary games for collaborative innovation, and diversifies enterprises' strategies in cooperation. Second, we consider project benefits across different cooperation levels from the perspective of unequal status between cooperating LEs and SMEs. This perspective enhances the understanding of cooperation among heterogeneous enterprises. Finally, through simulation, we examine the long-term and short-term effects of technology spillovers, government subsidies, and changes in project income on cooperation. Doing so provides actionable insights for policymakers and participating enterprises in innovation consortia to optimize management performance, thereby enhancing the output and benefits of innovation consortium R&D projects. Theoretically, this study's conclusions expand the boundaries of research on the stability of collaborative innovation in multi-strategy contexts. This research reveals the roles of subsidies and technology spillovers in maintaining cooperative stability. It provides new insights for governments in formulating effective innovation policies.

This paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes the modeling assumptions, parameters, and payoffs. Section 4 explains the procedure for constructing the evolutionary game model. Section 5 conducts numerical simulations, illustrating potential evolutionary equilibria. Section 6 analyzes parameter sensitivity through simulation. Section 7 presents our conclusions and managerial insights.

2. Literature review

2.1. Motivations affecting cooperation

Various motivations explain the development of an enterprise's competitive advantages. Influenced by opportunism, entrepreneurs strive for money, competitive advantage, and status, which leads to the use of extrinsic motivation for cooperation. This can result in issues such as free riding, which can undermine cooperation (Ferguson et al., 2020; Guzman et al., 2020). Alternatively, driven by internal factors like vision, trust, and social responsibility, enterprises engage in cooperation guided by intrinsic motivation, enhancing the value of the enterprise. Intrinsic motivation can generate creativity and be applied to product innovation (Henderson, 2021). Compared with economic incentives, intrinsic motivation demonstrates a superior capacity to deter speculation (Astola et al., 2022; Barrutia & Echebarria, 2021). While intrinsic and extrinsic motivations can coexist, it is noted that extrinsic motivation may hinder the creativity of R&D team members (Zhu et al., 2018). Most research studies classify motivation into internal and external types, with limited exploration of their coexistence.

2.2. Factors affecting motivation

Participants in the innovation consortium define their motivations by evaluating multiple factors. Previous research identifies factors that shape motivational preferences, including technology spillover (Grimpe et al., 2022; Hu et al., 2020). While spillover generates synergies, it can also induce speculative behavior in enterprises. In particular, when enterprises possess a high ability to imitate technology, concerns about free-riding in technological innovation can overshadow the positive externalities (Im & Shon, 2019). Variations in goals, budgets, and independent R&D capacities impact the knowledge-sharing processes of enterprises. Consequently, the leading enterprise consistently yields higher profits compared to its less advanced counterparts in cooperation (Majuri, 2022; Zheng et al., 2023). From the standpoint of technology spillover, suppliers are encouraged to engage in collaborative R&D to ensure that their benefits are competitive with other R&D models (Yang et al., 2024).

Project income plays a crucial role in determining motivation for cooperation, with the distribution of benefits significantly impacting the effort members invest (Hosseini-Motlagh et al., 2022). The expected income of the project is affected by the success rate of the project, which can change the cooperation strategy of the enterprise. Furthermore, subpar performance compared to expectations can dampen enterprises' enthusiasm for cooperation (Eggers & Kaul, 2018; Tian et al., 2021). Additionally, enterprises may benefit from the investments of other participants, potentially resulting in diminished individual efforts (Ning & Babich, 2018), or shirking. Government subsidies also have mixed effects. Moderate subsidies can foster innovation among project participants, whereas excessive subsidies can crowd out private investment (Li, Yi, & Zhao, 2020). Moreover, enterprises may not necessarily intensify their efforts upon receiving subsidies but instead may engage in free-riding behaviors instead (Song et al., 2022). Hence, project income, technology spillovers, and government subsidies stand as pivotal factors influencing cooperation. Our work investigates the combined effects of these multiple factors.

2.3. Evolutionary game models

Due to bounded rationality and imperfect information in inter-firm interactions, several scholars have developed evolutionary game models to evaluate the stability of cooperation. Participants in evolutionary games continually refine their strategies. After a prolonged period of evolution, the dynamic system may reach an evolutionary equilibrium, a stable state where the strategies of all participants remain constant. Simulations can forecast challenging outcomes in evolutionary games and enhance decision-making processes. The pure strategy within the game can evolve into an evolutionary stable strategy, which can be illustrated through simulation (Cheng et al., 2020; Oliveira et al., 2019). Some scholars have also explored the evolution of cooperation systems through simulations, primarily focusing on income. Consequently, extra income, costs, and policies influence the stability of cooperation. Elevating shared income to a specific threshold is essential for supporting cooperative synergy (Li, Cao, & Chi, 2020; Zhang et al., 2022). Liu et al. (2017) categorize cooperation strategies as positive or negative, based on the relationship between penalty intensity and cost within an evolutionary game. Zan et al. (2021) emphasize the constructive impact of government policies on the stability of collaborative innovation using an evolutionary game and simulations. They underscore the efficacy of incentive measures over penalties. Nevertheless, in specific projects like agricultural innovation projects, innovation benefits exert a more substantial influence on evolution than policies (Luo et al., 2023). In short, prior research has explored the factors influencing the stability of cooperation. However, there is an incomplete understanding of the mechanisms behind these factors, and further analysis is required based on project-specific characteristics.

Existing research refers to collaborations dominated by opportunism as “Opportunistic R&D collaboration”. It has been shown that subsidies in such collaborations have a dual effect on innovation efficiency (Dimakopoulou et al., 2023). However, the mechanism through which subsidies operate within the decision-making framework for R&D collaboration has not been fully explored. Therefore, we consider multi-strategy scenarios to provide additional insights into how incentive-based subsidies influence the evolution of collaboration among firms. Furthermore, by contrast with Luo et al. (2023), which views the government as a direct participant, we use the subsidy level to represent the government's regulatory role indirectly. This approach offers new perspectives for devising regulatory policies in a relaxed innovation environment for enterprises of differing statuses, thereby expanding the research boundaries of collaborative innovation. The literature closely related to this research is shown in Table 1.

To summarize, the research discussed above highlights the impact of various factors on corporate cooperation motivation, emphasizing the need for further study. Additionally, current applications of evolutionary games are confined to symmetric scenarios featuring equal participants and binary strategies. By contrast, this work uses an asymmetric three-strategy evolutionary game model and simulations to study the evolving dynamics of corporate motivations for cooperation within innovation consortia. It considers variables such as project income, technology spillovers, and government subsidies.

3. Model description

We provide a description of the game scenario considered. Within an innovation consortium, the LEs and the SMEs adopt initial motivations with respect to cooperation after evaluating internal and external factors. These motivations influence resource allocation and work effort, resulting in success rates and expected project income alterations. Consider two groups, one consisting of LEs and the other of SMEs, both participating in comparable innovation consortia within the same region. Within these groups, individual entities gradually mimic strategies that yield greater returns. Over time, this may result in the establishment of evolutionary stable strategies that most participants adopt. Our evolutionary game model identifies the evolutionary equilibria of motivations for cooperation and shows how motivation evolves. Fig. 1 provides an overview of the evolutionary model.

3.1. Assumptions

In order to study the evolution of cooperation, the model makes the following assumptions.

Assumption 1 (Strategic Options). In key technology projects, the optional strategies for LEs and SMEs within innovation consortia include intrinsic, extrinsic, and mixed motivations.

Intrinsic Motivation (IM): Driven by internal factors, enterprises choose IM for cooperation and cooperate actively. Their objective is to optimize cooperative business value, taking into consideration the interests of all stakeholders. Therefore, they actively engage with partners in the cooperation. IM has a positive impact on the project.

Extrinsic Motivation (EM): Driven by external factors such as increased self-interest, enterprises motivated by EM treat cooperation negatively. They prioritize self-interest and competitive advantage, often approaching the project negatively.

Mixed Motivation (MM): Enterprises adopt a strategy combining both intrinsic and extrinsic motivations. However, extrinsic motivation undermines intrinsic motivation's positive effects.

Assumption 2 (Project Income and Distribution). The expected project return rate is denoted by r , and the total project cost is c . If the project succeeds, it generates income represented as P ($P = rc$). When both participants choose intrinsic motivation, they actively engage and

Table 1
Literature on collaborative innovation games.

Game method	Players: Optional strategies	Technology spillovers	Subsidy	Income	Literature
Static game	Duopoly enterprises.	✓		✓	Zhao and Ju (2012)
Sequential game	An upstream firm and a downstream firm.	✓		✓	Zhou et al. (2020)
Sequential game	Leaders and followers.	✓			Chen et al. (2021)
Evolutionary game	Innovators in the network: Cooperation or noncooperation.			✓	Li, Cao, and Chi (2020)
Evolutionary game	Universities and enterprises: Positive cooperation or negative cooperation.	✓			Zan et al. (2021)
Evolutionary game	University and research institute: Positively cooperate or negatively cooperate.	✓	✓		Liu et al. (2017)
Evolutionary game	Government: Participate or nonparticipation. OFDI Firms: Negative diffusion or active diffusion. Domestic Firms: Active adoption or negative adoption.	✓	✓		Yang et al. (2023)
Evolutionary game	Agricultural enterprise and university: Whether to collaborate on innovation. Government: Whether to provide subsidies.		✓	✓	Luo et al. (2023)
Evolutionary game	Leading enterprises: Intrinsic, extrinsic, or mixed cooperation strategies. Small and medium-sized enterprises: Intrinsic, extrinsic, or mixed cooperation strategies.	✓	✓	✓	This work.

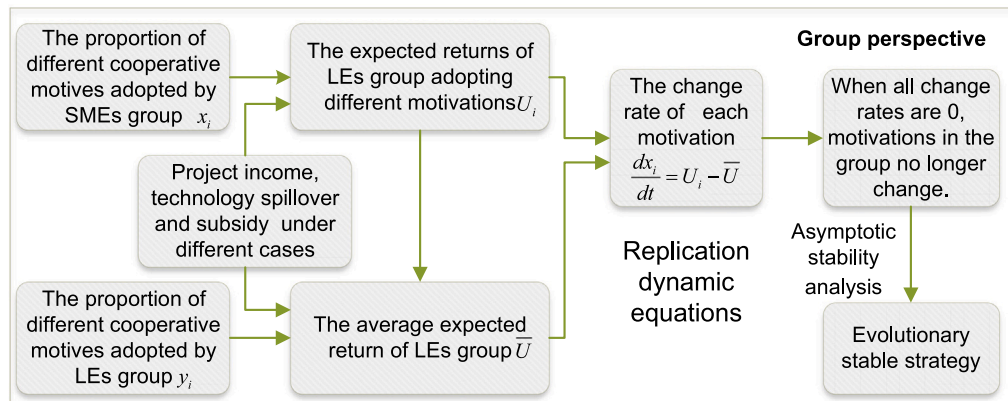


Fig. 1. Modeling components.

invest significantly in the project, resulting in high coordination. In this scenario, the project achieves a success rate of λ_1 , leading to project income I_1 ($I_1 = \lambda_1 P$). Alternatively, if the project experiences low coordination, it maintains a success rate of λ_2 , resulting in project income I_2 ($I_2 = \lambda_2 P$). The success rates are associated with varying coordination levels, satisfying: $0 < \lambda_2 < \lambda_1 \leq 1$. At the project's outset, both participants agreed on an income distribution ratio based on their respective investment costs. The LEs receive share k of the income, while the SMEs receive share $1 - k$. Given the dominant role of LEs in the project, the relevant range for k is limited to $[0.5, 1)$.

Assumption 3 (Subsidies and Speculative Income). Subsidies from the government are granted before project commencement. The amount of the subsidy is related to R&D costs. The proportion of cost-based subsidies is denoted as s , resulting in a total project subsidy of S ($S = sc$). These subsidies are allocated based on contractual proportions, with the LEs receiving kS and the SMEs receiving $(1 - k)S$. If enterprises opt for EM, subsidies are considered speculative gains, regardless of project success.

Assumption 4 (Technology Spillover). The LEs share their proprietary technology when driven by IM. Then, if the SMEs adopt IM or MM, they can access the advantages associated with technology spillover, denoted by T . We denote the technology acquisition capability of the SMEs as n . The income of technology spillover for SMEs can be written as $T = n(1 - k)c$. Nevertheless, this scenario presents a potential loss for the LEs. In another scenario, where the LEs adopt mixed motivation and the SMEs opt for IM, the SMEs can still accrue technology spillover benefits as T . In an innovation consortium, LEs typically consist of cross-industry enterprises that do not have competitive relationships. Meanwhile, SMEs do not share their core technologies with other SMEs as they primarily collaborate with LEs. Therefore, our research focuses on the technology spillover between SMEs and LEs.

Table 2 summarizes the model's parameters, which are all constrained to be positive values.

3.2. Payoffs

The project income varies with different combinations of motivations. The payoff matrix represents the payoff for each case. The extra

Table 2
Model parameters and descriptions.

Parameter	Value	Definition
λ_1	$0 < \lambda_1 \leq 1$	The expected success rate of the project under a high level of coordination.
λ_2	$0 < \lambda_2 < \lambda_1$	The expected success rate of the project under a low level of coordination.
r	$r > 0$	The expected return rate of the project.
c	$c > 0$	The total cost of the project.
n	$0 \leq n < 1$	The technology acquisition capability of the SMEs.
s	$0 < s < 1$	The proportion of government subsidies for projects.
I_1	$\lambda_1 P$	The total project income under a high level of coordination.
I_2	$\lambda_2 P$	The total project income under a low level of coordination.
T	$n(1-k)c$	The loss value of technology spillover to LEs. It is also the income value for the SMEs.
S	sc	The total amount of government subsidies to the project.
P	rc	The project income resulting from project success.
k	$0.5 \leq k < 1$	The income distribution ratio of the LEs in the project.

income discussed below represents supplementary project revenue from increased enterprise efforts. It is defined as $I_1 - I_2 = (\lambda_1 - \lambda_2)rc$.

Case 1: Both participating entities choose IM. Their robust cooperation leads to high project coordination and generates significant extra income. The LEs incur losses from technology spillovers, whereas the SMEs derive benefits from technology spillovers.

Case 2: The LEs choose IM, while the SMEs adopt MM. The LEs make substantial investments and contribute effort, while the SMEs make basic investments with speculative behavior. This results in poor coordination, leading to diminished project returns. Technology spillover gains and losses mirror those in Case 1.

Case 3: The LEs choose IM, while the SMEs adopt EM. The SMEs perceive projects negatively. Despite the absence of the SMEs' assistance, the LEs can struggle to achieve project success through internal R&D capabilities. Poor coordination levels lead to no gain or a loss from technology spillover.

Case 4: The LEs choose MM, while the SMEs adopt IM. Despite the SMEs' active cooperation, the LEs perceive them as outsourced entities. The LEs' investment covers only basic project operations, resulting in poor coordination. Technology spillover gains and losses are the same as in Case 1.

Case 5: Both participants choose MM. Funds and resources can only sustain the basic operation of the project, leading to limited coordination. Minimal communication reduces the influence of technology spillovers on gains and losses.

In **Cases 6–9**, project failures result from insufficient motivation or resources. Both participants can only obtain speculative incomes.

Case 6: The LEs choose MM, while the SMEs adopt EM. The LEs make minimal investments, utilizing subsidies to mitigate potential project failure, which leads to zero LE income. The SMEs acquire speculative income.

Case 7: The LEs choose EM, while the SMEs choose IM. The LEs' negative approach poses challenges for the SMEs to leverage their abilities. The LEs obtain speculative income equal to subsidies, while subsidies reimburse the SMEs for their project inputs.

Case 8: The LEs choose EM, while the SMEs adopt MM. The LEs' negative approach leads to a decrease in SMEs' investment, ultimately causing project failure. The LEs obtain speculative income equal to subsidies, while the SMEs have no income.

Case 9: Both participants choose EM, adopting a negative approach to the project and participating in subsidy fraud. Both participants obtain speculative gains equal to subsidies.

The payoff for each pair of strategic motivations is shown in Table 3. The proportions of the SMEs choosing intrinsic motivation (IM), mixed motivation (MM), and extrinsic motivation (EM) are defined as x_1 , x_2 , x_3 , where $x_3 = 1 - x_1 - x_2$. Similarly, the proportions of the LEs choosing IM, MM, and EM are defined as y_1 , y_2 , y_3 , where $y_3 = 1 - y_1 - y_2$.

4. Stability analysis of equilibrium points

The LE and SME groups adopt varying motivations within the same evolutionary game of cooperation motivation, resulting in differing profit outcomes. The linchpin of the evolutionary system is imitation. Within the dynamics of imitation, if a strategy's payoff surpasses the average payoff, more participants will adopt that strategy. Hence, superior strategies increasingly dominate while inferior ones gradually vanish. These dynamic systems may ultimately attain an equilibrium, drawing neighboring states toward it (Friedman, 1991). In this section, we provide the expression for the average return in the three-strategy game, present the dynamic replication equation that accounts for proportional strategy changes over time, and conduct a stability analysis of equilibrium points.

4.1. Replication dynamic equations of SMEs

For the SMEs, we present the calculation process for replicating dynamic equations. Given $x_3 = 1 - x_1 - x_2$ and $y_3 = 1 - y_1 - y_2$, the following four replication dynamic equations are sufficient to demonstrate the changes in the proportions of all the strategies in this game.

(1) The expected return is calculated as follows. Here, U_{SME1} , U_{SME2} , and U_{SME3} represent the expected returns associated with the SMEs choosing IM, MM, and EM, respectively.

$$\begin{aligned}
 U_{SME1} &= y_1((1-k)\lambda_1 rc + n(1-k)c) \\
 &\quad + y_2((1-k)\lambda_2 rc + n(1-k)c) \\
 U_{SME2} &= y_1((1-k)\lambda_2 rc + n(1-k)c) + y_2(1-k)\lambda_2 rc \\
 U_{SME3} &= y_1((1-k)\lambda_2 rc + (1-k)sc) + y_2((1-k)sc) \\
 &\quad + (1-y_1-y_2)(1-k)sc.
 \end{aligned}$$

(2) Then, the SMEs' average expected return ($\bar{U}_{SME}$) is calculated as follows.

$$\bar{U}_{SME} = x_1 U_{SME1} + x_2 U_{SME2} + (1-x_1-x_2) U_{SME3}.$$

(3) The replication dynamic equation of the SMEs choosing IM (g_1) is shown in Eq. (1).

$$\begin{aligned}
 g_1 &= \frac{dx_1}{dt} = x_1(U_{SME1} - \bar{U}_{SME}) \\
 &= x_1(y_1((1-k)\lambda_1 rc + n(1-k)c) + y_2((1-k)\lambda_2 rc + n(1-k)c) \\
 &\quad - x_1(y_1((1-k)\lambda_1 rc + n(1-k)c) + y_2((1-k)\lambda_2 rc + n(1-k)c)) \\
 &\quad - x_2(y_1((1-k)\lambda_2 rc + n(1-k)c) + y_2(1-k)\lambda_2 rc) \\
 &\quad - (1-x_1-x_2)(y_1((1-k)\lambda_2 rc + (1-k)sc) \\
 &\quad + y_2(1-k)sc + (1-y_1-y_2)(1-k)sc)).
 \end{aligned} \tag{1}$$

(4) The replication dynamic equation of the SMEs choosing MM (g_2) is shown in Eq. (2).

$$g_2 = \frac{dx_2}{dt} = x_2(U_{SME2} - \bar{U}_{SME}) \tag{2}$$

Table 3
Payoffs of the cooperative motivation game.

Case	LEs' strategies	SMEs' strategies	LEs' income	SMEs' income	Explanations
Case 1	IM	IM	$kI_1 - T$	$(1-k)I_1 + T$	Efficient collaboration led to high project returns. The presence of technology spillover (T) resulted in a loss to LEs and an income to SMEs.
Case 2	IM	MM	$kI_2 - T$	$(1-k)I_2 + T$	Compared to Case 1, the level of returns from collaboration has decreased to I_2 .
Case 3	IM	EM	kI_2	$(1-k)(I_2 + S)$	Both players gained project return. The subsidy s received by the SMEs was not used for the project and is considered speculative income.
Case 4	MM	IM	$kI_2 - T$	$(1-k)I_2 + T$	Compared to Case 1, the level of returns from collaboration has decreased to I_2 .
Case 5	MM	MM	kI_2	$(1-k)I_2$	Due to the lack of smooth cooperation, both players only gained project profits, with no technology spillover.
Case 6	MM	EM	0	$(1-k)S$	The project failed. The SMEs gained speculative profits.
Case 7	EM	IM	kS	0	The project failed. The LEs gained speculative profits.
Case 8	EM	MM	kS	0	The project failed. The LEs gained speculative profits.
Case 9	EM	EM	kS	$(1-k)S$	Both players colluded fraudulently to obtain subsidies as speculative gains.

Table 4
The general form of payoff matrix.

Players	SMEs			
	Strategies	IM(x_1)	MM(x_2)	EM(x_3)
LEs	IM(y_1)	a_{11}, b_{11}	a_{12}, b_{12}	a_{13}, b_{13}
	MM(y_2)	a_{21}, b_{21}	a_{22}, b_{22}	a_{23}, b_{23}
	EM(y_3)	a_{31}, b_{31}	a_{32}, b_{32}	a_{33}, b_{33}

$$\begin{aligned}
&= x_2 (y_1 ((1-k)\lambda_2 rc + n(1-k)c) + y_2 (1-k)\lambda_2 rc \\
&\quad - x_1 (y_1 ((1-k)\lambda_1 rc + n(1-k)c) + y_2 ((1-k)\lambda_2 rc + n(1-k)c)) \\
&\quad - x_2 (y_1 ((1-k)\lambda_2 rc + n(1-k)c) + y_2 (1-k)\lambda_2 rc) \\
&\quad - (1-x_1-x_2) (y_1 ((1-k)\lambda_2 rc + (1-k)sc) \\
&\quad \quad + y_2 (1-k)sc + (1-y_1-y_2) (1-k)sc).
\end{aligned}$$

The quantitative relations of the three replication dynamics equations are presented here for application to sensitivity analysis in the subsequent sections.

Proposition 1. Since $x_1 + x_2 + x_3 = 1$, $g_1 = x_1 (U_{SME1} - \bar{U}_{SME})$ and $g_2 = x_2 (U_{SME2} - \bar{U}_{SME})$, the replication dynamics equation of the SMEs choosing EM is $g_{EM}^{SME} = x_3 (U_{SME3} - \bar{U}_{SME}) = -g_1 - g_2$.

Proof. The participants of the game in this study are the leading enterprises (LEs) and small and medium-sized enterprises (SMEs). To extend the conclusion to a more general case, the general form of the payoff matrix of the two-player three-strategy game is shown in Table 4.

In the table, x_j and y_i are the proportions of different strategies adopted by SMEs and LEs respectively in the evolutionary process, a_{ij} represents the payoff of the LEs when they choose strategy i and the SMEs choose strategy j , and the payoff to the SMEs in the same case is b_{ij} . Strategy $i, j \in \{1, 2, 3\}$ represents the three strategies of IM, MM and EM, respectively (see Table 4). The expected returns to the SMEs when they adopt different strategies and the average expected returns to the SMEs are as follows:

$$U_{SME1} = y_1 b_{11} + y_2 b_{21} + y_3 b_{31} \quad (3)$$

$$U_{SME2} = y_1 b_{12} + y_2 b_{22} + y_3 b_{32} \quad (4)$$

$$U_{SME3} = y_1 b_{13} + y_2 b_{23} + y_3 b_{33} \quad (5)$$

$$\bar{U}_{SME} = x_1 U_{SME1} + x_2 U_{SME2} + x_3 U_{SME3}. \quad (6)$$

The replication dynamic equations of the SMEs when adopting different strategies are as follows:

$$g_{IM}^{SME} = x_1 (U_{SME1} - \bar{U}_{SME}) \quad (7)$$

$$g_{MM}^{SME} = x_2 (U_{SME2} - \bar{U}_{SME}) \quad (8)$$

$$g_{EM}^{SME} = x_3 (U_{SME3} - \bar{U}_{SME}). \quad (9)$$

By adding Eqs. (7), (8), and (9) together and substituting Eq. (6), we obtain:

$$\begin{aligned}
g_{IM}^{SME} + g_{MM}^{SME} + g_{EM}^{SME} &= x_1 U_{SME1} + x_2 U_{SME2} + x_3 U_{SME3} - (x_1 + x_2 + x_3) \bar{U}_{SME} \\
&= \bar{U}_{SME} - (x_1 + x_2 + x_3) \bar{U}_{SME} \\
&= 0 \quad \text{if } x_1 + x_2 + x_3 = 1.
\end{aligned} \quad (10)$$

Since x_1, x_2, x_3 represent the proportion of all strategies that can be taken by the SMEs and they cannot choose both strategies at the same time, $x_1 + x_2 + x_3 = 1$ must hold. Therefore, we derive $g_{EM}^{SME} = -g_{IM}^{SME} - g_{MM}^{SME}$, which corresponds to $g_{EM}^{SME} = -g_1 - g_2$ in Proposition 1. $\square$

4.2. Replication dynamic equations of LEs

(1) The expected return is calculated as follows. Here, U_{LE1} , U_{LE2} , and U_{LE3} represent the expected returns associated with the LEs choosing IM, MM, and EM, respectively.

$$\begin{aligned}
U_{LE1} &= x_1 (k\lambda_1 rc - n(1-k)c) + x_2 (k\lambda_2 rc - n(1-k)c) \\
&\quad + (1-x_1-x_2) (k\lambda_2 rc).
\end{aligned}$$

$$U_{LE2} = x_1 (k\lambda_2 rc - n(1-k)c) + x_2 (k\lambda_2 rc).$$

$$U_{LE3} = x_1 (ksc) + x_2 (ksc) + (1-x_1-x_2) (ksc).$$

(2) The LEs' average expected return is calculated as follows.

$$\bar{U}_{LE} = y_1 U_{LE1} + y_2 U_{LE2} + (1-y_1-y_2) U_{LE3}.$$

(3) The replication dynamic equation of the LEs choosing IM (g_3) is shown in Eq. (11). Also, the replication dynamic equation of the LEs choosing MM (g_4) is shown in Eq. (12).

$$\begin{aligned}
g_3 &= \frac{dy_1}{dt} = y_1 (U_{LE1} - \bar{U}_{LE}) \\
&= y_1 ((\lambda_1 rck - n(1-k)c) x_1 + (\lambda_2 rck - n(1-k)c) x_2 + k\lambda_2 rc (1-x_1-x_2) \\
&\quad - y_1 ((\lambda_1 rck - n(1-k)c) x_1 + (\lambda_2 rck - n(1-k)c) x_2 + k\lambda_2 rc (1-x_1-x_2)) \\
&\quad - y_2 ((\lambda_2 rck - n(1-k)c) x_1 + k\lambda_2 rc x_2) \\
&\quad - (1-y_1-y_2) (cksx_1 + cksx_2 + skc (1-x_1-x_2))).
\end{aligned} \quad (11)$$

$$\begin{aligned}
g_4 &= \frac{dy_2}{dt} = y_2(U_{LE2} - \bar{U}_{LE}) \\
&= y_2((\lambda_2 rck - n(1-k)c)x_1 + k\lambda_2 rcx_2 - y_1((\lambda_1 rck - n(1-k)c)x_1 \\
&\quad + (\lambda_2 rck - n(1-k)c)x_2 + k\lambda_2 rc(1-x_1-x_2)) \\
&\quad - y_2((\lambda_2 rck - n(1-k)c)x_1 + k\lambda_2 rcx_2) \\
&\quad - (1-y_1-y_2)(cksx_1 + cksx_2 + skc(1-x_1-x_2))).
\end{aligned} \quad (12)$$

The replication dynamics equation of the LEs choosing EM is g_{EM}^{LE} . Similar to the proof of Proposition 1, since $y_3 = 1 - y_1 - y_2$, $g_3 = y_1(U_{LE1} - \bar{U}_{LE})$ and $g_4 = y_2(U_{LE2} - \bar{U}_{LE})$, it can be derived that $g_{EM}^{LE} = -g_3 - g_4$.

4.3. Stability analysis results

According to Friedman (1991), a fixed point with local asymptotic stability is an evolutionary equilibrium. Fixed points are found by solving the system of differential Eqs. (13). The cooperative motivation game within the innovation consortium is inherently asymmetric, precluding the interchangeability of subjects. Within this asymmetric evolutionary game, the Nash equilibrium strict mixed strategy is not asymptotically stable (Hofbauer & Sigmund, 1998). Hence, we analyze whether the pure strategy constitutes a locally asymptotically stable point (LAS). Given $g_{EM}^{SME} = -g_1 - g_2$ and $g_{EM}^{LE} = -g_3 - g_4$, the stability analysis only requires simultaneous consideration of g_1, g_2, g_3 , and g_4 .

$$\begin{cases}
g_1 = \frac{dx_1}{dt} = x_1(U_{SME1} - \bar{U}_{SME}) = 0 \\
g_2 = \frac{dx_2}{dt} = x_2(U_{SME2} - \bar{U}_{SME}) = 0 \\
g_3 = \frac{dy_1}{dt} = y_1(U_{LE1} - \bar{U}_{LE}) = 0 \\
g_4 = \frac{dy_2}{dt} = y_2(U_{LE2} - \bar{U}_{LE}) = 0.
\end{cases} \quad (13)$$

The Lyapunov linear method employs the Jacobian matrix as the coefficient matrix to linearize the nonlinear system. The Jacobian matrix for this game is presented below.

$$Jacobian = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \frac{\partial g_1}{\partial x_2} & \frac{\partial g_1}{\partial y_1} & \frac{\partial g_1}{\partial y_2} \\ \frac{\partial g_2}{\partial x_1} & \frac{\partial g_2}{\partial x_2} & \frac{\partial g_2}{\partial y_1} & \frac{\partial g_2}{\partial y_2} \\ \frac{\partial g_3}{\partial x_1} & \frac{\partial g_3}{\partial x_2} & \frac{\partial g_3}{\partial y_1} & \frac{\partial g_3}{\partial y_2} \\ \frac{\partial g_4}{\partial x_1} & \frac{\partial g_4}{\partial x_2} & \frac{\partial g_4}{\partial y_1} & \frac{\partial g_4}{\partial y_2} \end{bmatrix}. \quad (14)$$

Lemma 2. If the real part of the eigenvalues of the coefficient matrix is negative, then the original nonlinear system possesses an asymptotically stable equilibrium point (Slotine, 1991).

Proposition 3. The fixed point $V_1(1, 0, 1, 0)$ is an evolutionary equilibrium if the following inequalities hold:

$$\begin{cases}
(\lambda_1 r + n - s)k - n > 0 \\
(\lambda_1 - \lambda_2)r + n - s > 0.
\end{cases} \quad (15)$$

Proof. For the dynamical system described by Eq. (13), the Jacobian matrix is derived based on Eq. (14). The fixed point $V_9(1, 0, 1, 0)$ represents $x_1 = y_1 = 1$ and $x_2 = x_3 = y_2 = y_3 = 0$. By substituting V_1 , the coefficient matrix (J_1) of the system at this fixed point is obtained:

$$J_1 = \begin{bmatrix} ((\lambda_1 - \lambda_2)r + n - s)(k-1)c & c(n-s)(k-1) & 0 & 0 \\ 0 & cr(k-1)(\lambda_1 - \lambda_2) & 0 & 0 \\ 0 & 0 & -c((r\lambda_1 + n - s)k - n) & -c((r\lambda_2 + n - s)k - n) \\ 0 & 0 & 0 & -ckr(\lambda_1 - \lambda_2) \end{bmatrix}.$$

Let z_i^j denote the J_i 's i th eigenvalue of the coefficient matrix. Solving the equation $\det(J_i - Z_i I) = 0$, the four eigenvalues of J_1 are

obtained as follows:

$$Z_1 = \begin{bmatrix} z_1^1 \\ z_2^1 \\ z_3^1 \\ z_4^1 \end{bmatrix} = \begin{bmatrix} -((\lambda_1 r + n - s)k - n)c \\ -ckr(\lambda_1 - \lambda_2) \\ -ckr(\lambda_1 - \lambda_2) \\ -((\lambda_1 - \lambda_2)r + n - s)(1-k)c \end{bmatrix}.$$

According to Assumption 2, we know that $\lambda_1 > \lambda_2$ and $k < 1$. Substituting these values into Z_1 , it is obtained that $z_2^1 < 0$ and $z_3^1 < 0$. Based on this, since $(\lambda_1 r + n - s)k - n > 0$, we have $z_1^1 < 0$. Since $(\lambda_1 - \lambda_2)r + n - s > 0$, $z_4^1 < 0$. It can be observed that all the eigenvalues are strictly negative. According to Lemma 2, under this condition, V_1 is a fixed point with locally asymptotically stability, meaning that V_1 serves as an evolutionary equilibrium. $\square$

Proposition 4. Fixed point $V_3(0, 0, 1, 0)$ is an evolutionary equilibrium if the following inequalities hold:

$$\begin{cases}
\lambda_2 r - s > 0 \\
(\lambda_1 - \lambda_2)r + n - s < 0.
\end{cases} \quad (16)$$

Proof. The fixed point $V_3(0, 0, 1, 0)$ represents $x_3 = y_1 = 1$ and $x_1 = x_2 = y_2 = y_3 = 0$. By following the same procedure as in the proof of Proposition 3, the coefficient matrix (J_3) of the system at the fixed point V_3 can be derived as:

$$J_3 = \begin{bmatrix} -((\lambda_2 - \lambda_1)r + n - s)(k-1)c & 0 & 0 & 0 \\ 0 & -c(n-s)(k-1) & 0 & 0 \\ 0 & 0 & -ck(r\lambda_2 - s) & ksc \\ 0 & 0 & 0 & -k\lambda_2 rc \end{bmatrix}.$$

Solving the equation $\det(J_i - Z_i I) = 0$, the four eigenvalues of J_3 are obtained as follows:

$$Z_3 = \begin{bmatrix} z_1^3 \\ z_2^3 \\ z_3^3 \\ z_4^3 \end{bmatrix} = \begin{bmatrix} -k\lambda_2 rc \\ -ck(\lambda_2 r - s) \\ c(n-s)(1-k) \\ ((\lambda_1 - \lambda_2)r + n - s)(1-k)c \end{bmatrix}.$$

Given that all parameters are non-negative, it follows that $z_1^3 = -k\lambda_2 rc < 0$. According to Assumption 2, we know that $\lambda_1 > \lambda_2$ and $k < 1$. Then, since $\lambda_2 r - s > 0$, $z_2^3 < 0$. Since $(\lambda_1 - \lambda_2)r + n - s < 0$, $z_4^3 < 0$. Based on the aforementioned conditions and $(\lambda_1 - \lambda_2)r + n - s < 0$, it can be concluded that $n - s < 0$ also holds true. Therefore, $z_3^3 < 0$. Since all the eigenvalues are strictly negative, according to Lemma 2, V_3 is a fixed point with locally asymptotically stability, meaning that V_3 serves as an evolutionary equilibrium. $\square$

Proposition 5. Fixed point $V_9(0, 0, 0, 0)$ is an evolutionary equilibrium if $\lambda_2 r - s < 0$.

Proof. $V_9(0, 0, 0, 0)$ represents $x_1 = x_2 = y_1 = y_2 = 0$ and $x_3 = y_3 = 1$. By following the same procedure as in the proof of Proposition 3, the coefficient matrix (J_9) of the system at the fixed point V_9 can be derived as:

$$J_9 = \begin{bmatrix} (k-1)sc & 0 & 0 & 0 \\ 0 & (k-1)sc & 0 & 0 \\ 0 & 0 & ck(r\lambda_2 - s) & 0 \\ 0 & 0 & 0 & -ksc \end{bmatrix}.$$

Solving the equation $\det(J_i - Z_i I) = 0$, the four eigenvalues of J_9 are obtained as follows:

$$Z_9 = \begin{bmatrix} z_1^9 \\ z_2^9 \\ z_3^9 \\ z_4^9 \end{bmatrix} = \begin{bmatrix} -ksc \\ (\lambda_2 r - s)kc \\ -(1-k)sc \\ -(1-k)sc \end{bmatrix}.$$

Table 5
Results of stability analysis.

Fixed point	Eigenvalues	Sign	Stability
$V_1 (1,0,1,0)$	$-\left((\lambda_1 r + n - s)k - n\right)c$ $cr(\lambda_1 - \lambda_2)(k - 1)$ $-ckr(\lambda_1 - \lambda_2)$ $\left((\lambda_1 - \lambda_2)r + n - s\right)(k - 1)c$	unsure – – unsure	Uncertain
$V_2 (0,1,1,0)$	$-n(k - 1)c$ $c(n - s)(k - 1)$ $-\left((\lambda_2 r + n - s)k - n\right)c$ $-cr(\lambda_1 - \lambda_2)(k - 1)$	+ unsure unsure +	Unstable point
$V_3 (0,0,1,0)$	$-k\lambda_2 rc$ $-ck(\lambda_2 r - s)$ $-c(n - s)(k - 1)$ $-\left((\lambda_1 - \lambda_2)r + n - s\right)(k - 1)c$	– unsure unsure unsure	Uncertain
$V_4 (1,0,0,1)$	$n(k - 1)c$ $-\left((\lambda_2 r + n - s)k - n\right)c$ $c(\lambda_2 r + n - s)(k - 1)$ $ckr(\lambda_1 - \lambda_2)$	– unsure unsure +	Unstable point
$V_5 (0,1,0,1)$	$n(k - 1)c$ $-n(k - 1)c$ $-ck(\lambda_2 r - s)$ $c(\lambda_2 r - s)(k - 1)$	– + unsure unsure	Unstable point
$V_6 (0,0,0,1)$	ksc $k\lambda_2 rc$ $-c(\lambda_2 r - s)(k - 1)$ $-\left((k - 1)\lambda_2 r - ks - n + s\right)c$	+ + unsure unsure	Unstable point
$V_7 (1,0,0,0)$	0 $-cs(k - 1)$ $ck(r\lambda_2 + n - s)$ $c(k(r\lambda_2 + n - s) - n)$	0 + unsure unsure	Unstable point
$V_8 (0,1,0,0)$	0 $-cs(k - 1)$ 0 0	0 + 0 0	Unstable point
$V_9 (0,0,0,0)$	$-ksc$ $c(\lambda_2 r - s)k$ $cs(k - 1)$ $cs(k - 1)$	– unsure – –	Uncertain

Given that all parameters are non-negative, it follows that $z_1^9 = -ksc < 0$. Based on [Assumption 2](#), we know that $k < 1$. By substituting this result into Z_9 , we consequently derive $z_3^9 < 0$ and $z_4^9 < 0$. Since $\lambda_2 r - s < 0$, it is obtained that $z_2^9 = (\lambda_2 r - s)kc < 0$. Since all the eigenvalues are strictly negative, according to [Lemma 2](#), V_9 is a fixed point with locally asymptotically stability, meaning that V_9 serves as an evolutionary equilibrium. $\square$

[Table 5](#) shows the asymptotic stability of each fixed point. The column “Sign” shows the sign of the real part of the eigenvalue, where “unsure” represents that the sign is uncertain.

Proposition 6. Only V_1 , V_3 , V_9 can be evolutionary equilibria in this game.

Proof. By [Lemma 2](#), the condition for each fixed point to become LAS is that the real parts of all eigenvalues are strictly negative. As shown in [Table 5](#), the eigenvalues associated with the remaining fixed points do not satisfy this condition. $\square$

5. Equilibrium condition derivation and evolutionary trajectory analysis

Simulation-based phase diagrams illustrate the evolutionary trajectories of the cooperative state and serve to validate the results of the stability analysis. [Figs. 2\(a\)](#) to [4\(a\)](#) depict the evolutionary outcomes of

strategies employed by the LEs and the SMEs. The data comes from [Department of Social, Science and Technology, and Cultural Statistics National Bureau of Statistics of the PRC \(2022a, 2022b\)](#). To represent the evolutionary trajectory in a unified three-dimensional image, [Fig. 2\(b\)](#) to [4\(b\)](#) display the evolutionary phase diagrams for x_1 , x_2 in combination with y_i , $i = 1, 2, 3$. Thirty initial points were generated randomly to represent diverse initial conditions. The iteration count is set at 30,000, a value determined through experimentation. This count is sufficient to ensure that each initial point reaches an evolutionary equilibrium. For ease of interpretation, we mark the trajectory’s evolution direction, starting point, and end point.

The value of sensitivity analysis in [Section 6](#) is based on [Section 5](#). This research uses the ratio of the R&D intensity of LEs and SMEs. Then, we solve the following equation to obtain the value of k .

$$\frac{k}{1 - k} = \frac{\text{LEs' R\&D intensity}}{\text{SMEs' R\&D intensity}}$$

5.1. The evolutionary equilibrium of collaborating actively

Based on the conditions outlined in [Proposition 3](#) for V_1 to achieve an evolutionary equilibrium, we set $c = 1$, $r = 15.0\%$, $k = 67.0\%$, $n = 1.0\%$, $\lambda_1 = 25.5\%$, $\lambda_2 = 17.0\%$, $s = 2.0\%$. [Fig. 2](#) displays the evolution phase diagram for this scenario.

Both LEs and SMEs progressively increase their proportion of IM, ultimately reaching 100%, which means both parties are collaborating

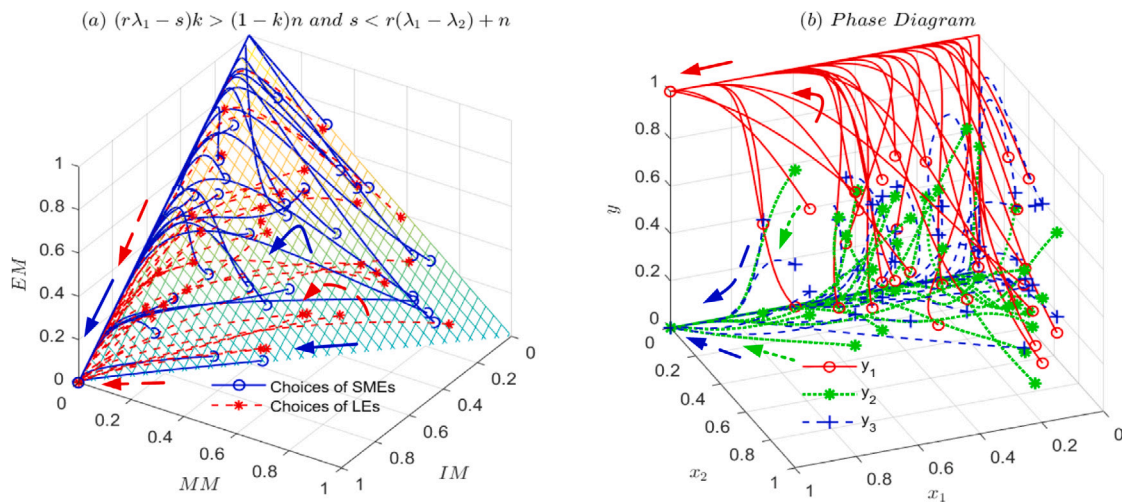


Fig. 2. The evolutionary equilibrium of collaborating actively.

actively in this equilibrium. Nevertheless, trajectories corresponding to certain initial strategies reveal that certain SMEs initially adopt EM strategies, transitioning to IM after a sufficient duration. This phenomenon becomes more apparent when the initial strategy contains more speculative strategies. Fig. 2(b) illustrates the trajectories of both parties adopting IM exhibits a spiral pattern. Ultimately, only the strategy combination of jointly adopting IM can persist in the game. This confirms that the V_1 is an evolutionary equilibrium.

Proposition 7. *The factors guiding active cooperation differ between the LEs and the SMEs. The LEs prioritize net income from high-level cooperation. The SMEs focus on the income gap between high-level and poor-level cooperation, which equals the income from additional efforts.*

Proof. Multiply both sides of the first inequality in Proposition 3 by $c(c > 0)$, followed by rearranging the terms. Similarly, multiply both sides of the second inequality by $(1 - k)c$ and rearrange its terms as well. Subsequently, incorporate the parameter relationships from Table 2 to derive the following results:

$$\begin{aligned} (\lambda_1 r + n - s)k - n > 0 &\Rightarrow k\lambda_1 rc - (1 - k)nc > ksc \\ &\Rightarrow kI_1 - T > kS. \end{aligned} \quad (17)$$

$$\begin{aligned} (\lambda_1 - \lambda_2)r + n - s > 0 &\Rightarrow (1 - k)(\lambda_1 - \lambda_2)rc + (1 - k)nc > (1 - k)sc \\ &\Rightarrow (1 - k)(I_1 - I_2) + T > (1 - k)S. \end{aligned} \quad (18)$$

Eq. (17) is the tradeoff faced by the LEs between effort and technology spillover. The LEs choose IM and actively engage in projects when the net income under high-level cooperation is higher than the subsidy. Eq. (18) represents the tradeoff for the SMEs. Given that this research views subsidies as speculative income, the SMEs choose IM when the combined extra income and technology spillover income surpasses the speculative income. In Eq. (18), $((1 - k)(I_1 - I_2))$ represents the additional project income attained by SMEs through their efforts. Also, the benefits of technological spillover come from additional efforts. Summarizing, the income from additional efforts can encourage the SMEs to engage in active cooperation. $\square$

5.2. The evolutionary equilibrium of free-riding in SMEs

Based on the conditions outlined in Proposition 4 for V_3 to achieve an evolutionary equilibrium, we set $c = 1$, $r = 15.0\%$, $k = 67.0\%$, $n = 1.0\%$, $\lambda_1 = 25.5\%$, $\lambda_2 = 17.0\%$, $s = 2.5\%$. In such a scenario, the SMEs approach the project negatively and derive project income from the efforts of the LEs, which is free-riding. Fig. 3 displays the evolutionary phase diagram for this scenario.

From Fig. 3, we observe that the LEs ultimately embrace IM, while the SMEs choose EM. The evolutionary trajectory presents a spiral shape. Intriguingly, when the proportion of mixed motivation in SMEs' initial strategy is high, they will initially shift toward IM before transitioning to EM. Conversely, with a high initial proportion of MM for the LEs, they initially shift toward EM before transitioning to IM. The LEs tend to increase the degree of speculation first and then consider cooperating actively. In short, the speculative behavior of the SMEs lags in this equilibrium. The SMEs engage in active cooperation before considering speculation.

Proposition 8. *High subsidies have a crowding-out effect. When the income from subsidies surpasses that from the additional efforts of the SMEs, the SMEs will turn to adopting EM strategies, resulting in the equilibrium of free-riding in SMEs.*

Proof. Adopting the same procedures as those in Proposition 7, we analyze the conditions in Proposition 4 that render V_3 an evolutionary equilibrium. The reasons for the emergence of this equilibrium are derived as follows:

$$\lambda_2 r - s > 0 \Rightarrow \lambda_2 rc > sc \Rightarrow I_2 > S. \quad (19)$$

$$(\lambda_1 - \lambda_2)r + n - s < 0 \Rightarrow (1 - k)(I_1 - I_2) + T < (1 - k)S. \quad (20)$$

Eq. (19) indicates that the expected project income under poor coordination must exceed the subsidy treated as speculative income. Eq. (20) has the opposite sign compared to Eq. (18). This shows that the project income from extra efforts and technology spillover by the SMEs must be less than subsidies. Summarizing, if speculative gains surpass the extra income generated by SMEs' extra efforts, they will reduce their efforts on the project. This reflects the crowding-out effect of high subsidies. $\square$

5.3. The evolutionary equilibrium of joint speculation

Based on the conditions outlined in Proposition 5 for V_9 to achieve an evolutionary equilibrium, we set $c = 1$, $r = 15.0\%$, $k = 67.0\%$, $n = 1.0\%$, $\lambda_1 = 25.5\%$, $\lambda_2 = 17.0\%$, $s = 10.0\%$. Fig. 4 illustrates that both the LEs and the SMEs steadily increase their adoption of EM, ultimately reaching 100%. This corresponds to a situation where enterprises conspire to commit subsidy fraud. Unlike the other two equilibria, where the evolution trajectories are spiral, the evolutionary trajectories of this equilibrium evolves directly toward the equilibrium point of joint speculation, regardless of the initial proportion.

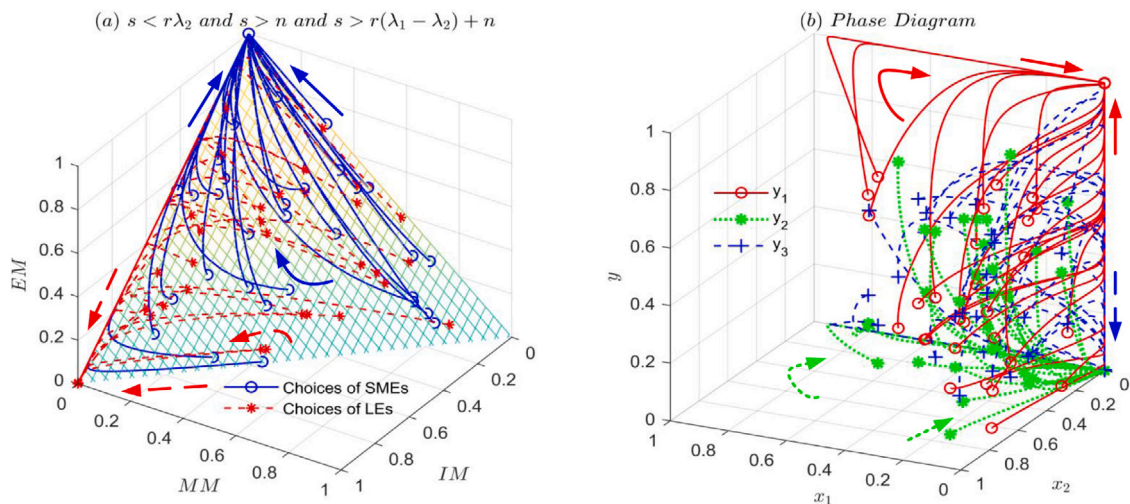


Fig. 3. The evolutionary equilibrium of free riding in SMEs.

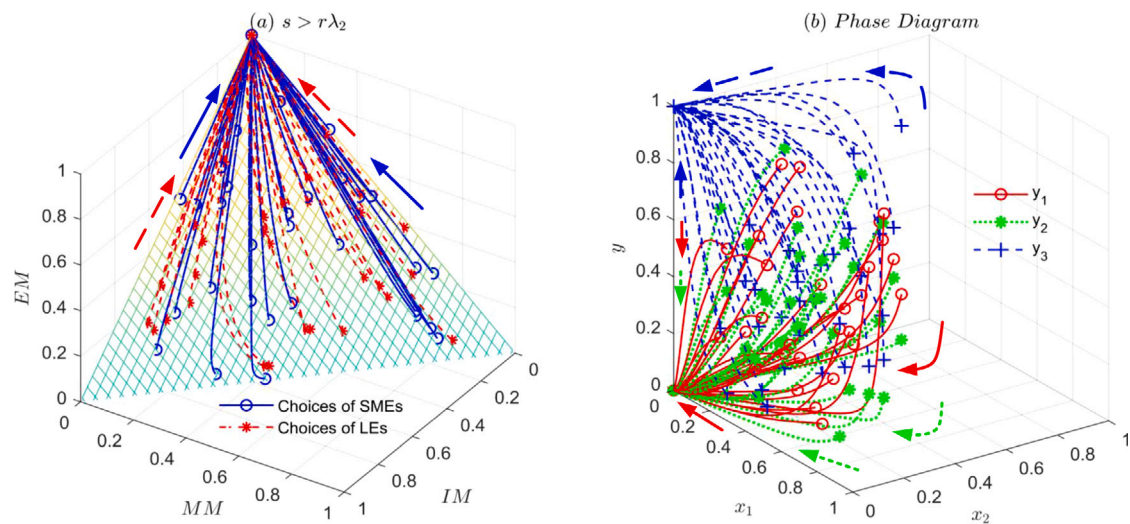


Fig. 4. The evolutionary equilibrium of joint speculation.

Proposition 9. If the expected return of poor-level cooperation is lower than the subsidy, both sides collaborate to engage in speculation to defraud government subsidies.

Proof. Similar to the derivation of Eq. (19), the condition for this equilibrium is reorganized as follows:

$$\lambda_2 r - s < 0 \Rightarrow \lambda_2 r c < s c \Rightarrow I_2 < S. \quad (21)$$

From the perspective of the project's overall income, both participants will resort to speculation to secure subsidies when cooperative income under poor coordination falls short of subsidies. $\square$

We also solve the Nash equilibria of this problem in a static game framework, which again generates the same three equilibrium points. However, the assumptions and implications of the solutions differ between the two scenarios. In a static game, both players make optimal choices simultaneously under full rationality. Static game Nash equilibria do not necessarily have local asymptotic stability. In contrast, the cooperative state within an innovation consortium is a dynamic system. Enterprises typically exhibit bounded rationality, and cooperation strategies change over time. An evolutionary game is therefore more suitable for this problem, yielding equilibrium solutions that satisfy the players. Evolutionary equilibria emphasize that strategies in

the neighborhood of the equilibrium point eventually evolve towards it, making both the process of change and the equilibrium outcome important in practice. Further, based on the evolutionary trajectory, recommendations for management from a temporal perspective can be provided. Additionally, Friedman (1998) shows that a Nash equilibrium is a necessary condition for an evolutionary equilibrium. This confirms the correctness of our solutions.

6. Sensitivity analysis

According to the goal of the innovation consortium, V_1 is the state of cooperation desired by the government and other stakeholders. Based on Section 5, the participants in innovation consortia and governments can formulate measures to make V_1 become the only evolutionary equilibrium. Consequently, we conduct sensitivity analysis under the condition that V_1 becomes the sole equilibrium. Additionally, we show that the corresponding evolutionary equilibrium points are attained in the three cases above through the previous simulation, irrespective of the initial strategy proportions of both sides. Hence, we consider an initial strategy distribution of 1/3 for each strategy. The initial values of the parameters are set to: $r = 0.15, n = 0.01, \lambda_1 = 0.255, \lambda_2 = 0.17, s = 0.02, k = 0.67$. In each parameter adjustment, only one variable's value is modified, while the others remain constant.

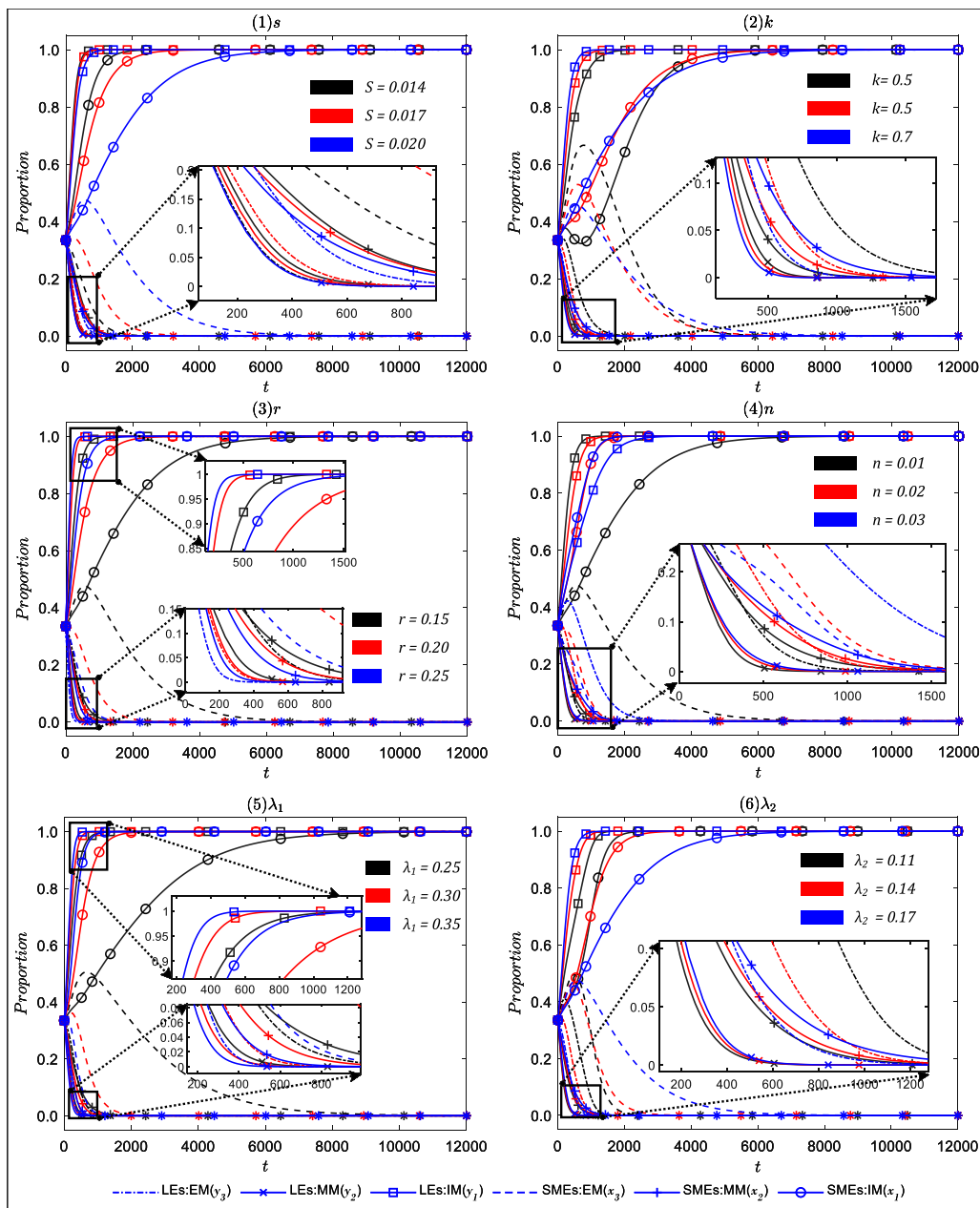


Fig. 5. Sensitivity analysis results.

Given that other parameters depend on the cost (c), we only examine the sensitivity of $s, k, r, n, \lambda_1, \lambda_2$ while setting $c = 1$, representing the per-unit cost.

6.1. The proportion of government subsidies for projects

The proportion of government subsidy (s) varies by region, contingent upon the subsidy intensity of local governments. According to incentive policies for high-tech enterprises in cities with the highest innovation index in China, such as Shanghai, Shenzhen, and Hangzhou, subsidies are typically less than 5% of R&D spending. Hence, we set $s \in \{0.014, 0.017, 0.020\}$. Sensitivity analysis results are shown in Fig. 5(1). The change in the subsidy ratio noticeably affects the SMEs' motivation but has minimal impact on the LEs. As the proportion of subsidies increases, the time for SMEs' strategies to converge towards IM extends. Additionally, the marginal effect is incremental. Consequently, it can be

concluded that subsidies hold greater appeal for SMEs for speculative purposes.

6.2. The income distribution ratio of LEs

The income distribution ratio of the LEs (k) is determined through mutual agreement. Given the dominant position of the LEs, we set $k \in \{0.50, 0.60, 0.70\}$. Fig. 5(2) shows the sensitivity analysis results. As k increases, the convergence speed of the LEs toward adopting IM strategies accelerates. For the SMEs, an increase in the income of the LEs corresponds to a decrease in the SMEs' income. Surprisingly, it also leads to a decrease in the proportion of the SMEs adopting EM. Since project returns are distributed proportionally, the common factor $(1 - k)$ can be derived from each replication dynamic equation of the SMEs. This indicates that an increase in k reduces the speed of strategic imitation among the SMEs. During the strategy evolution process, the LEs rapidly converge to adopting the IM strategy, whereas the adoption

rate of the EM strategy by the SMEs grows slowly. Consequently, early adoption of the IM strategy by the SMEs can lead to higher returns, thereby reducing the peak proportion of the EM strategy adopted by the SMEs throughout the evolutionary process.

6.3. The return rate of the project

The project's return rate (r) varies significantly depending on the industry involved in key technology development. To ensure comparable returns for similar projects in the evolutionary game, we set the value of r regarding the average cost-profit rate of each industry. The cost-profit rate is calculated as the ratio of total profit to total cost. According to data from [National Bureau of Statistics of China \(2022\)](#), we set $r \in \{0.15, 0.20, 0.25\}$. Fig. 5(3) shows the sensitivity analysis results. An increase in r accelerates the adoption of IM by both participants. From the detailed figure, as project returns increase, the LEs are more likely to abandon the EM strategy quickly. The SMEs exhibit greater sensitivity to this change. An increase in r can significantly boost the proportion of SMEs adopting IM during the early stages of evolution.

6.4. The technology acquisition capability of SMEs

The SMEs' technology acquisition capability (n) is measured by the percentage of their innovation costs covered by external R&D funding. For the LEs, this represents a loss due to technology spillover within the project, whereas it constitutes a profit for the SMEs. The value of n is 1.7% for medium-sized enterprises and 0.7% for small enterprises ([Department of Social, Science and Technology, and Cultural Statistics National Bureau of Statistics of the PRC, 2022b](#)). Hence, we set $n \in \{0.01, 0.02, 0.03\}$. The sensitivity analysis results are presented in Fig. 5(4). As n increases, there is an increase in the proportion of SMEs choosing IM, accompanied by a decrease in those choosing EM. Conversely, for the LEs, the time required for the strategy to converge towards IM increases as n grows. Additionally, its marginal effect exhibits an incremental trend, indicating that the LEs are vigilant about the potential damage resulting from technology spillovers.

6.5. The expected success rate of the project under a high coordination level

The expected success rate of the project under a high level of coordination (λ_1) is contingent on the R&D foundation and cooperative capabilities of both participating entities. In 2021, the proportion of enterprises engaging in R&D activities in China to achieve product innovation was 35.8% ([National Bureau of Statistics of China, 2022](#)). Hence, we set $\lambda_1 \in \{0.25, 0.30, 0.35\}$. The sensitivity analysis results are presented in Fig. 5(5). An increase in λ_1 results in a quicker convergence of both participants' strategies toward IM. Also, it reduces the proportion of the SMEs transitioning to EM in the early stages of evolution.

6.6. The project's expected success rate under a poor coordination level

The expected success rate of the project under a poor level of coordination (λ_2) primarily hinges on the industry and the independent R&D capabilities of the participating entities. Notably, it is influenced by the R&D capabilities of LEs within innovation consortia. This factor impacts the expected project return when cooperation is insufficient. Based on the cooperation development data of Chinese enterprises, ([Department of Social, Science and Technology, and Cultural Statistics National Bureau of Statistics of the PRC, 2022b](#)), we set $\lambda_2 \in \{0.11, 0.14, 0.17\}$. Fig. 5(6) shows that the impact of increasing λ_2 has contrasting effects on the evolution of both participants' strategies. Specifically, it accelerates the evolution time of the LEs adopting IM totally, with a diminishing marginal effect. Conversely, as the value of λ_2 increases, SMEs' strategies take more time to converge to IM. While the proportion of SMEs

using EM decreases in the early stages of evolution, it takes longer for them to reach the equilibrium point where EM is entirely absent.

To summarize, our sensitivity analysis yields the following insights:

1. High subsidies are prone to induce speculation in SMEs, inhibiting their active cooperation.
2. The increase in the income distribution ratio of LEs promotes the active cooperation of SMEs. It also reduces the proportion of SMEs using EM to speculate in the initial stage of cooperation.
3. A high project rate of return can stimulate active cooperation among enterprises, and SMEs display increased sensitivity to it.
4. LEs are particularly sensitive to the technology acquisition ability of SMEs. The SMEs are more sensitive to parameter changes than the LEs and exhibit a greater preference for the EM strategy compared to the MM strategy.
5. Enhancing the income from high-level cooperation expedites the innovation consortium's transition to active cooperation.
6. Ensuring the low-level cooperation income is a double-edged sword. While it prompts rapid transformation of the LEs towards active cooperation, it also extends the persistence of the speculative strategy of SMEs, which is detrimental to cooperation in the initial stages of the project.

7. Conclusions

This research introduces a multi-strategy evolutionary game model for innovation consortia. The primary objective of the work is to investigate the evolution of stable strategies for both LEs and SMEs within a multi-factor and multi-strategy framework. The research seeks to uncover variations in the responses of diverse enterprises to changing factors. Our contributions to the literature and to practice are outlined below.

7.1. Theoretical implications

First, the scenario in which both participants engage in cooperation driven by IM can become an evolutionary equilibrium. This finding aligns with the conclusions drawn from a dual-strategy evolutionary game about collaborative innovation ([Zan et al., 2021](#)). However, we further discover that this evolutionary equilibrium is not exclusive within a multi-strategy selection framework. Unilateral speculation by SMEs and joint speculation by both participating entities can also result in evolutionary equilibria. The emergence of different equilibria results from the distinct project income experienced by the two entities when evaluating technology spillovers and subsidies. LEs prioritize project income from high-level cooperation, whereas SMEs focus on the income gap across varying cooperation levels. In the case of poor underlying returns from the project, both entities are inclined to engage in speculation quickly. These research outcomes underscore the significance of enhancing project returns during high-level cooperation for the stability of collaborative efforts, thereby improving understanding of the evolution of multi-strategy collaborative innovation.

Second, it is surprising to discover that an increase in the revenue distribution ratio of the LEs can encourage the SMEs to adopt IM and reduce SME speculation in the early stages of cooperation. With the rise in the revenue share of LEs, the income of cooperation and free-riding concurrently decrease for SMEs. As technology spillover benefits remain unaffected by the distribution ratio, they become more appealing to SMEs compared to other benefits, encouraging active cooperation on their part. Similarly, the adverse effects from technology spillovers appear relatively minimal compared to the gains for the LEs, further fostering active cooperation. These research outcomes underscore the significance of augmenting the revenue distribution ratio of LEs and its correlation with technology spillovers. This also establishes a theoretical foundation for securing the dominant position of leading companies in collaborative research within innovation consortia.

Finally, SMEs exhibit greater sensitivity to changes in factors such as subsidies and rates of return, and their speculative behavior entails a time lag. In contrast, LEs primarily display sensitivity exclusively to technology spillovers. Our research provides additional evidence that high subsidies can impede innovative cooperation, highlighting the incremental impact of subsidy augmentation in inducing corporate speculation. Even within the subsidy ratio range where both participants eventually engage in active cooperation, higher subsidies correspond to prolonged evolution time for both participants, increasing the probability of speculation in the early stages of cooperation. These research findings unveil the differential impact of cooperation factors on heterogeneous enterprises in collaborative innovation research, thereby broadening the research perspective over long-term and short-term horizons.

7.2. Managerial and regulatory implications

First, our results suggest that governments should design differentiated and graduated subsidies based on an enterprise's historical R&D investment as well as the industry of the R&D technology. Considering the impact of subsidies on the stable outcomes and evolutionary process of innovation consortia, it is evident that the government should closely control the subsidy proportion to foster active cooperation. The government can accurately determine the subsidy ratio based on the historical R&D investment and innovative achievements of the participating enterprises, avoiding excessive subsidies. In practice, most local governments classify innovation subsidies into two levels, while cities with strong innovation capabilities subdivide their subsidy proportion into three or more levels for greater precision. For example, according to Beijing's 2020 subsidy policy for SMEs, SMEs with low R&D expenses are eligible for subsidies of no more than 5%. In contrast, for SMEs with R&D expenditures exceeding \$680,000, a subsidy cap of approximately \$27,000 is stipulated. Additionally, we recommend that the government consider the R&D success rates of different industries when adjusting subsidy rates, since the expected success rate of a project also impacts the stability of the cooperation. Considering that high-return projects stimulate positive cooperation among enterprises, projects in low-return industries require supplementary policy support.

Second, governments need to strengthen the post-evaluation of collaborative projects and reward high-level collaborations based on the evaluation results to encourage active cooperation among enterprises within innovation consortia. Enterprises that achieve high-quality R&D outcomes could be given media coverage and tax reductions, thereby expanding their influence and promoting an increase in their market value. For example, the company Xiamen Yingqu established a Metaverse Technology Innovation Consortium, which quickly achieved breakthroughs in several key technologies related to Metaverse scenarios within just over a year. The government awarded prizes and bonuses, promoting these accomplishments through various media channels. These incentives effectively fostered stable collaboration and attracted additional enterprises to join the consortium. Further, rewards should be customized for projects with different degrees of cooperation to prevent unilateral speculation by SMEs. Governments can facilitate the transformation of project outcomes, guarantee fundamental benefits post-project success, and deter collaborative speculation among enterprises.

Finally, for enterprises, we recommend agreeing on a higher revenue ratio of LEs to balance the effects of technology spillover and promote stable cooperation within the innovation consortium. This not only alleviates the concerns of LEs regarding technology spillover but also accentuates the advantages of technology spillover for SMEs. At the founding of an innovation consortium, the contract should explicitly define the high-income distribution for LEs. Additionally, due to SMEs' heightened sensitivity to factors like income and subsidies, they are prone to speculation. In practice, pilot operation schemes for innovation consortia in provinces such as Guangxi and Shanghai

recommend that cooperative benefits be distributed based on contributions, and that the ownership and distribution of intellectual property rights be clearly defined. These schemes require regular evaluations of the consortia, allowing for dynamic adjustments to membership based on the performance of participating enterprises. Our research provides theoretical support for the necessity of such mechanisms. We call on leading enterprises, as the drivers of cooperation, to establish project progress control mechanisms and improve mid-course exit mechanisms during the collaboration process to enable timely intervention for the delayed speculative tendencies of SMEs.

7.3. Future research opportunities

Our work identifies several important future research opportunities. First, this study primarily examines the dynamic evolution of cooperation between LEs and SMEs within innovation consortia, with government regulation quantified by subsidy intensity. Future research can involve the government as a participant, developing a three-party, multi-strategy evolutionary game model to analyze how different regulatory policies impact collaboration among stakeholders in these consortia. Second, enterprises involved in these consortia may be influenced by external conditions such as financial and trade environments. Introducing random perturbation terms into the model can enhance its applicability under uncertainty. Finally, for simplicity, this study does not account for the time value of money in relation to income. Future studies can explore various revenue distribution models and variations in discount factors to assess their implications for long-term cooperation.

CRedit authorship contribution statement

Zheng Yang: Validation, Methodology, Investigation, Writing – original draft, Formal analysis. **Lin Li:** Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis. **Nicholas G. Hall:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis. **Yongzeng Lai:** Supervision, Investigation, Formal analysis, Conceptualization, Writing – original draft. **Yijiang Zhou:** Writing – original draft, Validation, Methodology, Data curation.

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